Upload this 13-sheet PDF only. Expected answers are supplied separately; they are not drawing inputs.

AL NAHDA DESIGN HUB

Two-storey office fit-out / irregular footprint / double-height reception

Synthetic coordination and takeoff test - not an approved or buildable design.

Sheet	Content
G000	Scope, register and measurement rules
A101	Ground floor architecture - 1:100
A102	First floor architecture - 1:100
A201	Sections, stair and opening details
A301	Floor finishes and room register
A302	Ceiling finishes and heights
A401	Door instance schedule
A402	Window instance schedule
A403	Wall finishes and waterproofing
M101	HVAC measured routes and terminals
P101	Plumbing measured routes and fixtures
E101	Electrical equipment and cable routes
Q001	Scope checklist and calculation conventions

MEASUREMENT CONTRACT

1. Coordinates and dimensions are in metres. Grid dimensions are wall centreline (CL).
2. External CMU: 200 mm. Internal P01: 150 mm. GL01: full-height glass.
3. Room clear faces offset 100 mm from external CL; 75 mm from internal CL.
4. Partition lengths use full grid CL lengths. Do not deduct T-junction overlaps.
5. Quantities are net: deduct every scheduled door/window opening, regardless of size.
6. Plaster/paint are clear room-face quantities. Measure both faces of solid internal walls.
7. Opening jambs, reveals, sills and soffit returns are EXCLUDED from plaster/paint.
8. Floor / ceiling quantities use clear room polygons; not gross CL areas.
9. No floor or ceiling at F01. Reception roof-soffit ceiling belongs to G01 only.
10. Deduct 3 x 6 m stair zone from both core room floors; measure stair finishes separately.
11. All routes are net CL installed lengths. Add specified vertical lengths only.
12. Include no unspecified branches, allowances, fittings, reinforcement or contingency.
13. Room/fixture schedules repeat the plans: ONE item, not an additional copy.
14. Price using your own catalog. This fixture defines quantities, not market prices.

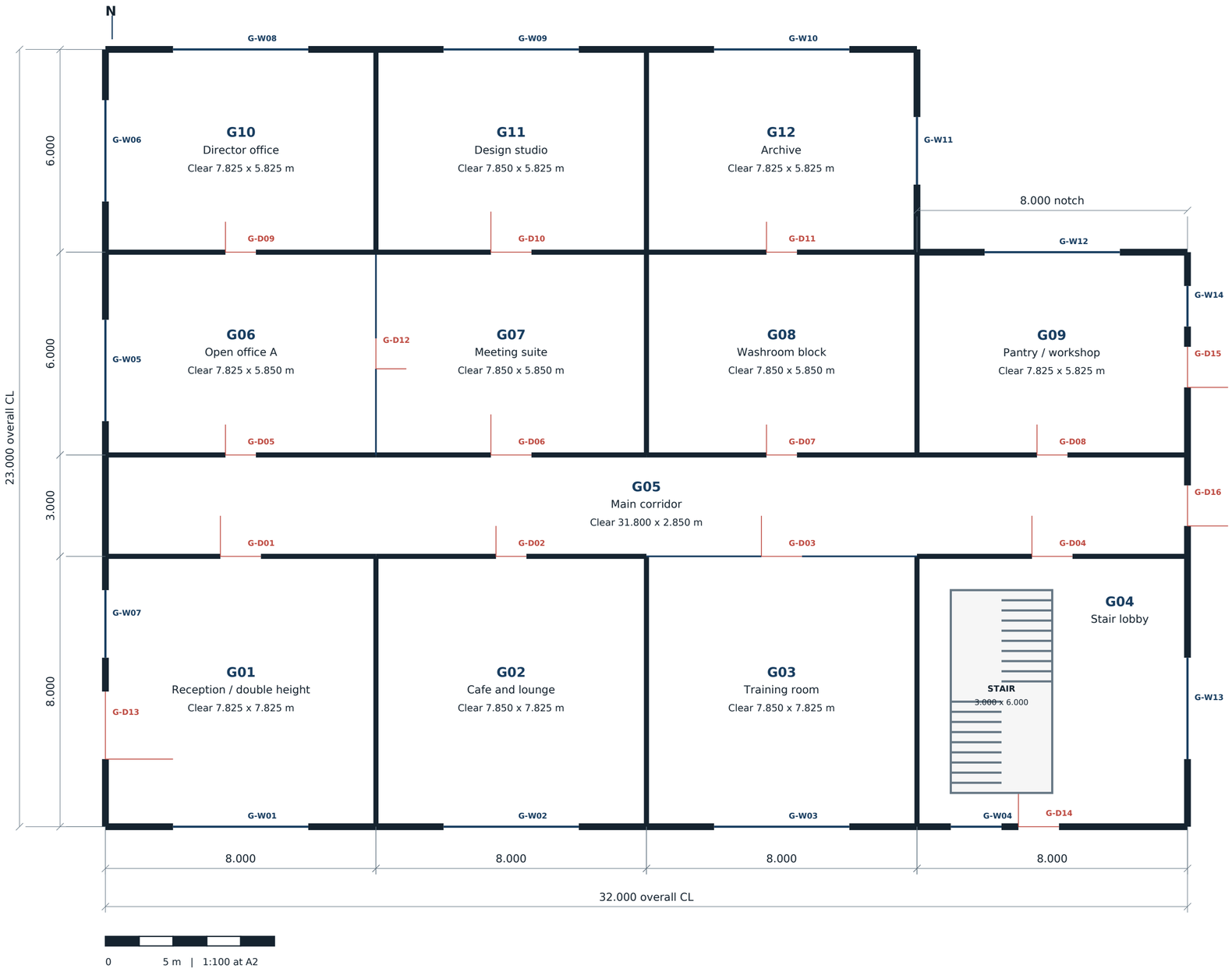
SCOPE: fit-out CMU/glazing, plaster, finishes, doors/windows, screed, waterproofing, stair finishes, scheduled HVAC/plumbing/electrical.

EXCLUDE: foundations, concrete frame/slabs, reinforcement, roof build-up, external works, demolition, lifts, sanitary cubicle partitions, furniture, joinery, branch services not drawn, duct fittings, pipe fittings, fire alarm cabling and all specialist design certification.

The building envelope and CMU are test quantities; geometry is deliberately coordinated by the measurement rules above.

Do not infer concrete or steel quantities from architectural floor area.

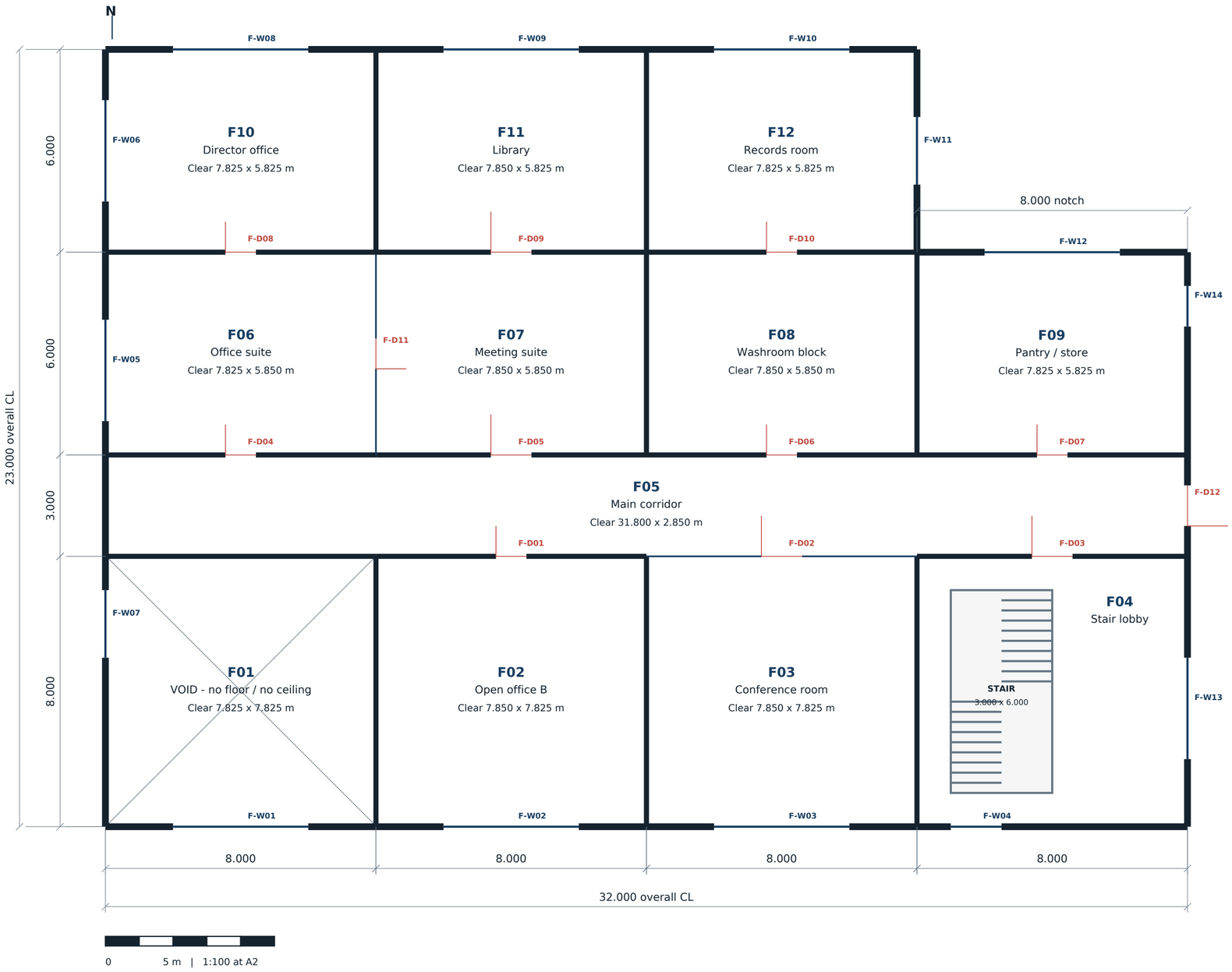
Scale 1:100 at original A2. Opening IDs are unique instances. Door/window sizes and locations: A401 / A402.



PLAN KEY / SETTING OUT

CL polygon: (0,0) (32,0) (32,17) (24,17) (24,23) (0,23).
Grid X: 0 / 8 / 16 / 24 / 32.
Grid Y: 0 / 8 / 11 / 17 / 23.
Solid external wall E01 = 200 mm CMU.
Solid internal wall P01 = 150 mm CMU.
GL01: full-height glass on the complete G/F03-to-G/F05 boundary, and G/F06-to-G/F07 boundary.
Glass openings DG09 / DG12 are measured separately as doors; deduct them from fixed glass.
Opening coordinates are CL endpoints, not room-face offsets.
Ground FFL: +0.000. First FFL: +4.200.
Stair zone: X25.000-28.000 / Y1.000-7.000.
Ground: exclude stair footprint from core floor. First: exclude stair opening.
F01 is a double-height VOID over reception: no upper floor or separate ceiling.

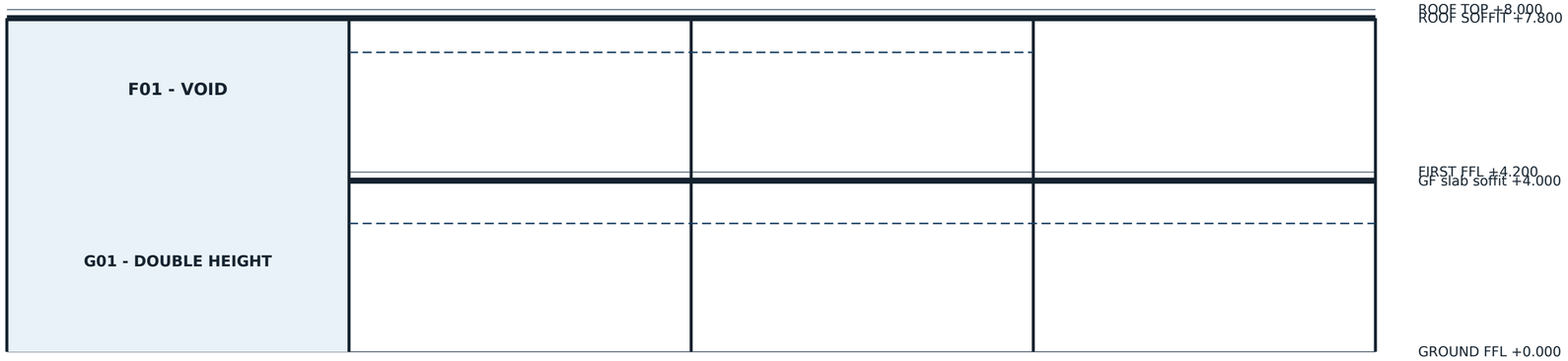
Scale 1:100 at original A2. Opening IDs are unique instances. Door/window sizes and locations: A401 / A402.



PLAN KEY / SETTING OUT

CL polygon: (0,0) (32,0) (32,17) (24,17) (24,23) (0,23).
Grid X: 0 / 8 / 16 / 24 / 32.
Grid Y: 0 / 8 / 11 / 17 / 23.
Solid external wall E01 = 200 mm CMU.
Solid internal wall P01 = 150 mm CMU.
GL01: full-height glass on the complete G/F03-to-G/F05 boundary, and G/F06-to-G/F07 boundary.
Glass openings DG09 / DG12 are measured separately as doors; deduct them from fixed glass.
Opening coordinates are CL endpoints, not room-face offsets.
Ground FFL: +0.000. First FFL: +4.200.
Stair zone: X25.000-28.000 / Y1.000-7.000.
Ground: exclude stair footprint from core floor. First: exclude stair opening.
F01 is a double-height VOID over reception: no upper floor or separate ceiling.

Orthographic test sections. Dimensions and stated quantities govern; no structural reinforcement scope.



SECTION A-A / Y=4.000 / 1:100

Level / component	Measured height / thickness
Ground external and internal walls	4.000 m above ground FFL
First external and internal walls	3.600 m above first FFL
Ground C01 / C02 ceiling	3.000 m above ground FFL
First C01 / C02 ceiling	2.800 m above first FFL
G01 C03 roof soffit	7.800 m above ground FFL; count once
G04 / F04 C03 slab soffit	4.000 m / 3.600 m above local FFL
Floor screed SC01	0.050 m over net floor finish areas only
Wet waterproofing WP01	Floor + 0.300 m wall upstand, separately
Stair ST01	20 risers x 0.210 m = 4.200 m
Stair finish treads	18 treads x 1.500 m wide x 0.300 m going
Stair finish risers	20 risers x 1.500 m wide x 0.210 m
Stair finish landings	2 landings x 1.500 x 3.000 m
Stair handrail HR01	2 rails x 6.000 m; no extra allowance
First void guardrail GR01	3 segments: 6.000 + 6.000 + 3.000 m

STAIR / FLOOR EXCLUSIONS

The 18.000 m2 rectangle on A101/A102 is the stair footprint/opening deduction.

Landing finishes are included only under ST01; do not include them again in core floor F03.

No extra skirting on the opening edge. Core skirting follows its four room walls only.

Screed under stair treads/risers/landings is excluded.

Rail lengths are measured segments specified here, not stair rise-derived lengths.

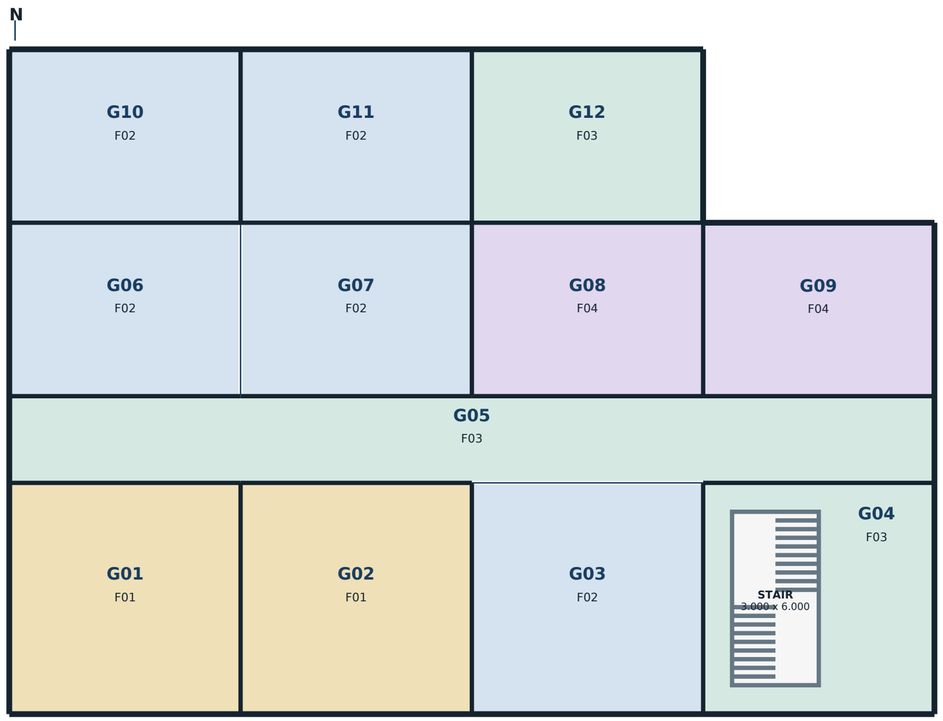
No upper F01 floor, screed, skirting, fittings or separate ceiling.

First-level walls around F01 remain full-height; no door opens into the void.

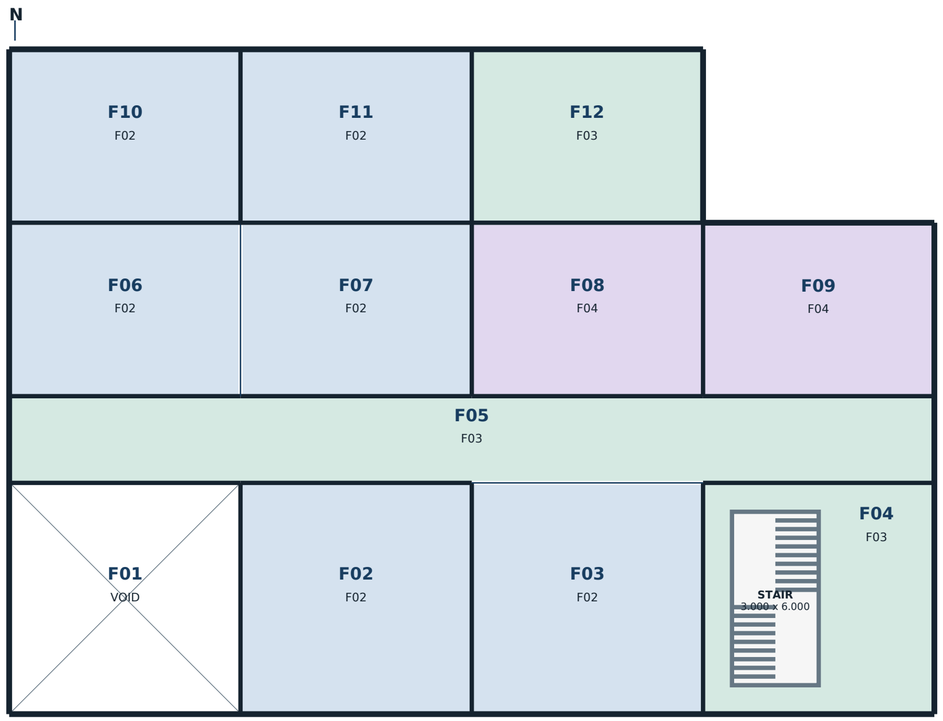
All internal glass is full local wall height, including above suspended ceilings.

Two floor finish plans at 1:150. Floor boundaries follow clear wall faces. Colours identify finish codes, not extra areas.

GROUND



FIRST



Code	Floor finish	Skirting
F01	600 x 600 porcelain floor	100 mm porcelain skirting
F02	LVT plank floor	100 mm painted MDF skirting
F03	600 x 600 terrazzo floor	100 mm terrazzo skirting
F04	Anti-slip ceramic wet floor	No skirting; full-height wall tile WT01

SC01 screed: 50 mm on every net floor finish area; exclude upper void and both 18 m2 stair zones.

Tile / LVT quantities are net installed area. Add no waste to measured BOQ quantities.

Skirting: clear room perimeter minus each door width. Do NOT deduct windows.

No skirting along full GL01 glass faces. No skirting in G08/G09/F08/F09 wet rooms.

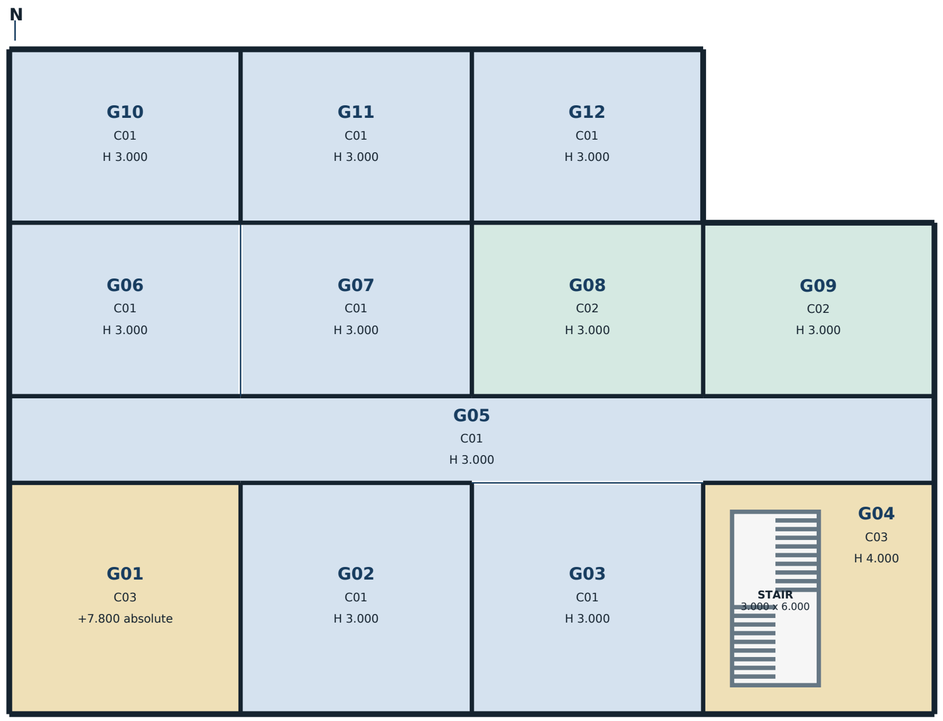
Skirting height does not multiply its measured linear quantity.

GF room	Floor	Ceiling	FF room	Floor	Ceiling
G01	F01	C03	F01	VOID	VOID
G02	F01	C01	F02	F02	C01
G03	F02	C01	F03	F02	C01
G04	F03	C03	F04	F03	C03
G05	F03	C01	F05	F03	C01
G06	F02	C01	F06	F02	C01
G07	F02	C01	F07	F02	C01
G08	F04	C02	F08	F04	C02
G09	F04	C02	F09	F04	C02
G10	F02	C01	F10	F02	C01
G11	F02	C01	F11	F02	C01
G12	F03	C01	F12	F03	C01

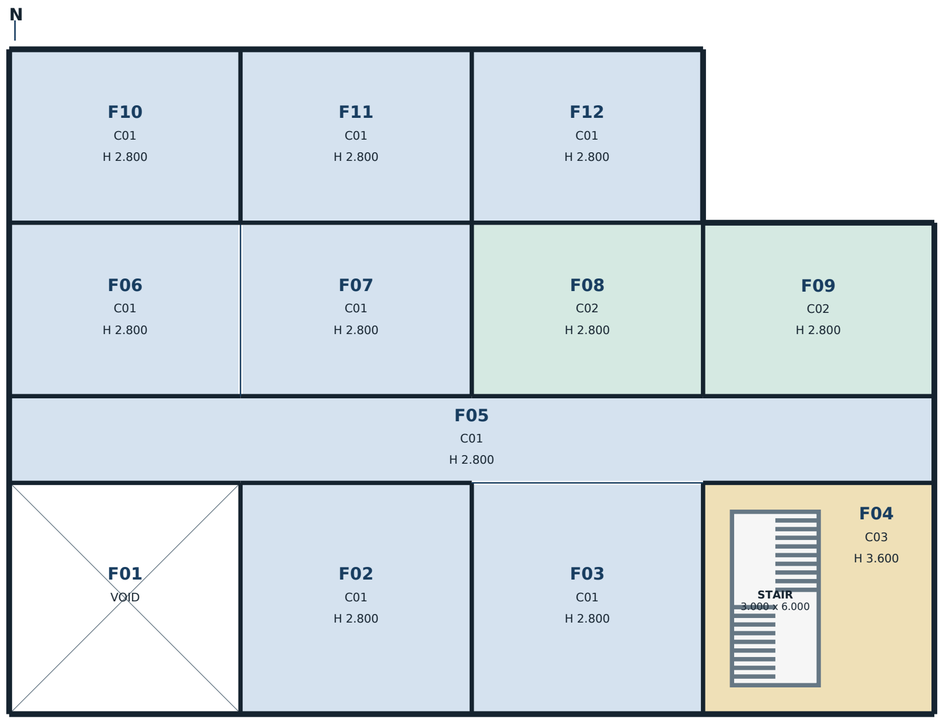
Units: metres unless noted. Dimensions govern. Do not measure a screenshot. Structural design / construction use excluded.

Plans at 1:150. Ceiling areas are net clear room areas, with the same core stair-zone deductions as floors.

GROUND



FIRST



Code	Description	Finish / measurement
C01	Suspended gypsum board ceiling	Net area; include one coat-system ceiling paint CP01
C02	Moisture-resistant gypsum ceiling	Net area; include one coat-system moisture paint CP02
C03	Exposed concrete soffit, painted	Net area only; no gypsum board. G01 belongs to roof above void.
VOID	F01 upper void	No separate ceiling: count its roof soffit under G01 only.

Deduct 18 m2 from G04 and F04 ceiling C03 for this benchmark; stairwell soffit / sloping stair undersides are excluded.

G01 C03 soffit is at +7.800. G04 C03 is H4.000. F04 C03 is H3.600. C01/C02 heights are on A201.

Do not deduct small lighting/diffuser penetrations from ceiling areas. No bulkheads, coves or ceiling-perimeter trims in scope.

All heights 2.400 m. Opening endpoints use the architectural CL system. Each ID appears once in the BOQ.

Instance	Type	W m	H m	Spaces	CL endpoints (x,y) m
G-D01	D02	1.200	2.400	G01 / G05	(3.4, 8) to (4.6, 8)
G-D02	D01	0.900	2.400	G02 / G05	(11.55, 8) to (12.45, 8)
G-D03	DG12	1.200	2.400	G03 / G05	(19.4, 8) to (20.6, 8)
G-D04	D02	1.200	2.400	G04 / G05	(27.4, 8) to (28.6, 8)
G-D05	D01	0.900	2.400	G06 / G05	(3.55, 11) to (4.45, 11)
G-D06	D02	1.200	2.400	G07 / G05	(11.4, 11) to (12.6, 11)
G-D07	D01	0.900	2.400	G08 / G05	(19.55, 11) to (20.45, 11)
G-D08	D01	0.900	2.400	G09 / G05	(27.55, 11) to (28.45, 11)
G-D09	D01	0.900	2.400	G10 / G06	(3.55, 17) to (4.45, 17)
G-D10	D02	1.200	2.400	G11 / G07	(11.4, 17) to (12.6, 17)
G-D11	D01	0.900	2.400	G12 / G08	(19.55, 17) to (20.45, 17)
G-D12	DG09	0.900	2.400	G06 / G07	(8, 13.55) to (8, 14.45)
G-D13	DE20	2.000	2.400	G01 / EXT	(0, 2) to (0, 4)
G-D14	DE12	1.200	2.400	G04 / EXT	(27, 0) to (28.2, 0)
G-D15	DE12	1.200	2.400	G09 / EXT	(32, 13) to (32, 14.2)
G-D16	DE12	1.200	2.400	G05 / EXT	(32, 8.9) to (32, 10.1)
F-D01	D01	0.900	2.400	F02 / F05	(11.55, 8) to (12.45, 8)
F-D02	DG12	1.200	2.400	F03 / F05	(19.4, 8) to (20.6, 8)
F-D03	D02	1.200	2.400	F04 / F05	(27.4, 8) to (28.6, 8)
F-D04	D01	0.900	2.400	F06 / F05	(3.55, 11) to (4.45, 11)
F-D05	D02	1.200	2.400	F07 / F05	(11.4, 11) to (12.6, 11)
F-D06	D01	0.900	2.400	F08 / F05	(19.55, 11) to (20.45, 11)
F-D07	D01	0.900	2.400	F09 / F05	(27.55, 11) to (28.45, 11)
F-D08	D01	0.900	2.400	F10 / F06	(3.55, 17) to (4.45, 17)
F-D09	D02	1.200	2.400	F11 / F07	(11.4, 17) to (12.6, 17)
F-D10	D01	0.900	2.400	F12 / F08	(19.55, 17) to (20.45, 17)
F-D11	DG09	0.900	2.400	F06 / F07	(8, 13.55) to (8, 14.45)
F-D12	DE12	1.200	2.400	F05 / EXT	(32, 8.9) to (32, 10.1)

D01: solid timber door 0.900 x 2.400. D02: solid timber double/accessible door 1.200 x 2.400.

DG09: glass door 0.900 x 2.400. DG12: glass door 1.200 x 2.400. Deduct from fixed GL01 glazing only.

DE12: external metal door 1.200 x 2.400. DE20: external double glazed door 2.000 x 2.400.

Door BOQ unit: number of complete door sets. Width x height is opening deduction, not quantity of door sets.

Solid doors in P01 deduct from CMU and both wall finishes. Exterior doors deduct from E01 and interior/exterior faces.

Net opening dimensions. Window BOQ measured as area by type; counts retained for checks. No window reveal finish scope.

Instance	Type	W m	H m	Sill m	Room	CL endpoints (x,y) m
G-W01	W01	4.000	2.400	0.900	G01	(2, 0) to (6, 0)
G-W02	W02	4.000	1.800	0.900	G02	(10, 0) to (14, 0)
G-W03	W02	4.000	1.800	0.900	G03	(18, 0) to (22, 0)
G-W04	W03	1.500	1.200	0.900	G04	(25, 0) to (26.5, 0)
G-W05	W04	3.000	1.800	0.900	G06	(0, 12) to (0, 15)
G-W06	W04	3.000	1.800	0.900	G10	(0, 18.5) to (0, 21.5)
G-W07	W05	2.000	1.800	0.900	G01	(0, 5) to (0, 7)
G-W08	W02	4.000	1.800	0.900	G10	(2, 23) to (6, 23)
G-W09	W02	4.000	1.800	0.900	G11	(10, 23) to (14, 23)
G-W10	W02	4.000	1.800	0.900	G12	(18, 23) to (22, 23)
G-W11	W05	2.000	1.800	0.900	G12	(24, 19) to (24, 21)
G-W12	W02	4.000	1.800	0.900	G09	(26, 17) to (30, 17)
G-W13	W04	3.000	1.800	0.900	G04	(32, 2) to (32, 5)
G-W14	W06	1.200	0.600	1.800	G09	(32, 14.8) to (32, 16)
F-W01	W11	4.000	1.500	0.900	F01	(2, 0) to (6, 0)
F-W02	W12	4.000	1.500	0.900	F02	(10, 0) to (14, 0)
F-W03	W12	4.000	1.500	0.900	F03	(18, 0) to (22, 0)
F-W04	W13	1.500	1.200	0.900	F04	(25, 0) to (26.5, 0)
F-W05	W14	3.000	1.500	0.900	F06	(0, 12) to (0, 15)
F-W06	W14	3.000	1.500	0.900	F10	(0, 18.5) to (0, 21.5)
F-W07	W15	2.000	1.500	0.900	F01	(0, 5) to (0, 7)
F-W08	W12	4.000	1.500	0.900	F10	(2, 23) to (6, 23)
F-W09	W12	4.000	1.500	0.900	F11	(10, 23) to (14, 23)
F-W10	W12	4.000	1.500	0.900	F12	(18, 23) to (22, 23)
F-W11	W15	2.000	1.500	0.900	F12	(24, 19) to (24, 21)
F-W12	W12	4.000	1.500	0.900	F09	(26, 17) to (30, 17)
F-W13	W14	3.000	1.500	0.900	F04	(32, 2) to (32, 5)
F-W14	W16	1.200	0.600	1.800	F09	(32, 14.8) to (32, 16)

Powder-coated aluminium double-glazed windows, measured by net scheduled opening area.

W01-W06 are ground types; W11-W16 are first-floor types. Different heights must not be merged incorrectly.

W06/W16 are high-level wet/service windows. Opening deductions apply even though smaller than 1 m2.

There is no minimum opening deduction threshold in this test. Window heads are sill + scheduled height.

Use room clear dimensions and opening schedules. Wall finish zones are not multiplied by coats.

Code	Material	Measurement rule
E01	200 mm external CMU	Net CL wall length x local height, less all external openings; BOQ m2
P01	150 mm internal CMU	Net CL wall length x local height, less solid door openings; BOQ m2
GL01	Full-height fixed internal glass	Net CL length x local height less DG09/DG12 glass door openings; BOQ m2
PL01	Internal plaster	All solid clear room wall faces, less openings; includes under wet wall tiles
WPNT	Internal emulsion paint	PL01 net faces minus WT01 tiled faces; includes above wet wall tiles
EPNT	Exterior weather paint	Net E01 wall area; no parapets or reveals
WT01	Wet-room ceramic wall tile	G08/G09/F08/F09: all four clear walls to 2.400 m; less doors/windows below 2.400
WP01	Wet floor waterproofing	G08/G09/F08/F09 clear floor area only
WP02	Waterproofing upstand	0.300 m high x wet-room clear perimeter minus door widths; BOQ m2

CLEAR-FACE WALL METHOD

For each room, measure 2 x (clear width + clear depth) x local wall height.

Subtract the entire clear length of any full-height GL01 face. Then deduct each opening in that room face once.

A solid internal door belongs to two rooms, so it is deducted once from each of its two finished faces.

For WT01, each door opens from floor to 2.400 m. High-level windows deduct only their overlap with 0-2.400 m.

G09 W06 and F09 W16: sill 1.800 m, height 0.600 m; full window area lies inside the tiled band.

G09 W02 / F09 W12: sill 0.900 m, heights 1.800 / 1.500; tile deduction clips at 2.400 m.

No WT01 in the other rooms. No tile skirting in wet rooms. Plaster remains included underneath tiles.

G01 local wall height is 4.000 m. F01 adds the upper 3.600 m wall faces around the same void.

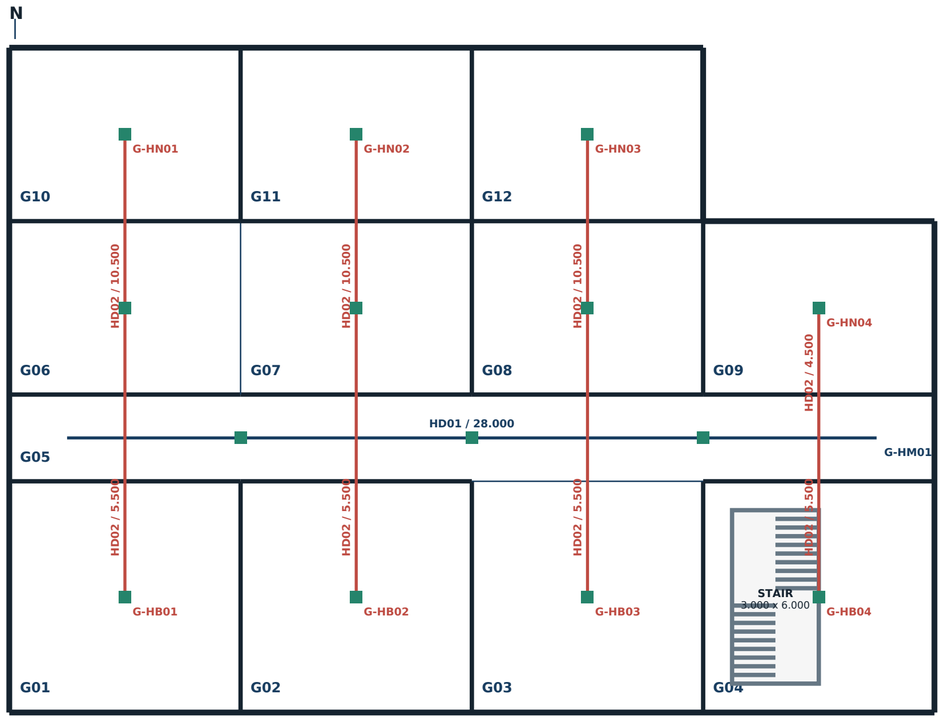
Suspended ceilings do not reduce partition heights or painted wall areas in this benchmark.

E01 and P01 quantities are area, not volume. If your BOQ uses m3: convert E01 x0.200 and P01 x0.150.

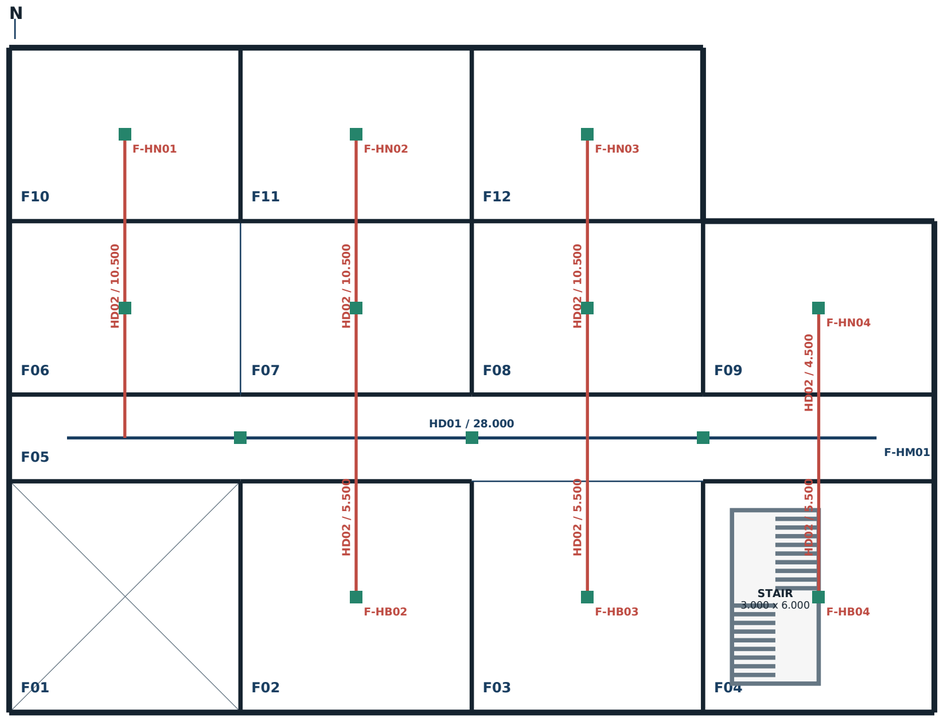
This is a declared test convention. Do not silently replace it with another standard measurement rule.

Plans 1:150. Route labels give each straight segment length. No fitting equivalents or hidden branch allowances.

GROUND



FIRST



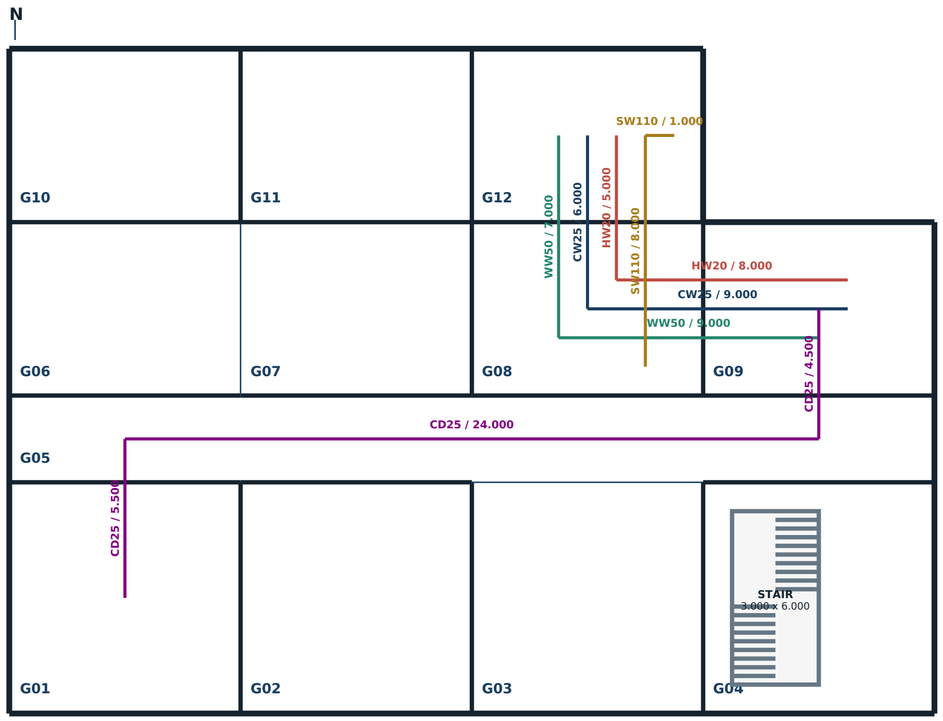
Code	Item	Specification / quantity rule
HD01	Main supply duct	800 x 400 mm GI; length CL. Sheet-metal area = 2 x (0.800+0.400) x length.
HD02	Branch supply duct	400 x 250 mm GI; length CL. Sheet-metal area = 2 x (0.400+0.250) x length.
HI01	External duct insulation	25 mm: use SAME nominal sheet-metal areas as HD01+HD02, no thickness expansion.
SD01	Supply diffusers	1 per occupied room except corridor: 3 each floor. No diffuser in F01.
RG01	Return grilles	1 per occupied room except G08/G09/F08/F09; 2 per corridor. No grille in F01.
EF01	Extract fans	2 per wet room, both levels. Ductwork for extract excluded.
FCU01	Fan-coil units	4 per level; scope is unit count only.

G-HB01 has an additional 4.800 m VERTICAL HD02 riser to the G01 high-level diffuser. All other duct verticals = zero.

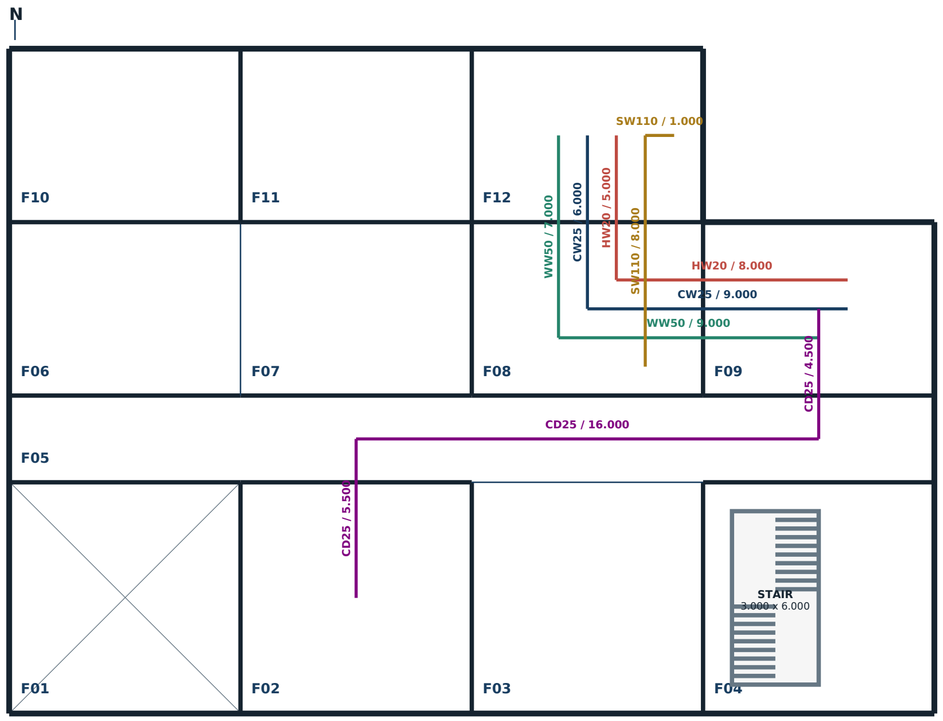
Diffusers/grilles/fans/FCUs above are a unique equipment schedule, not extra sets in addition to symbols.

Plans 1:150. Measured runs only; fixtures include installation, but no additional branch pipe allowance.

GROUND



FIRST



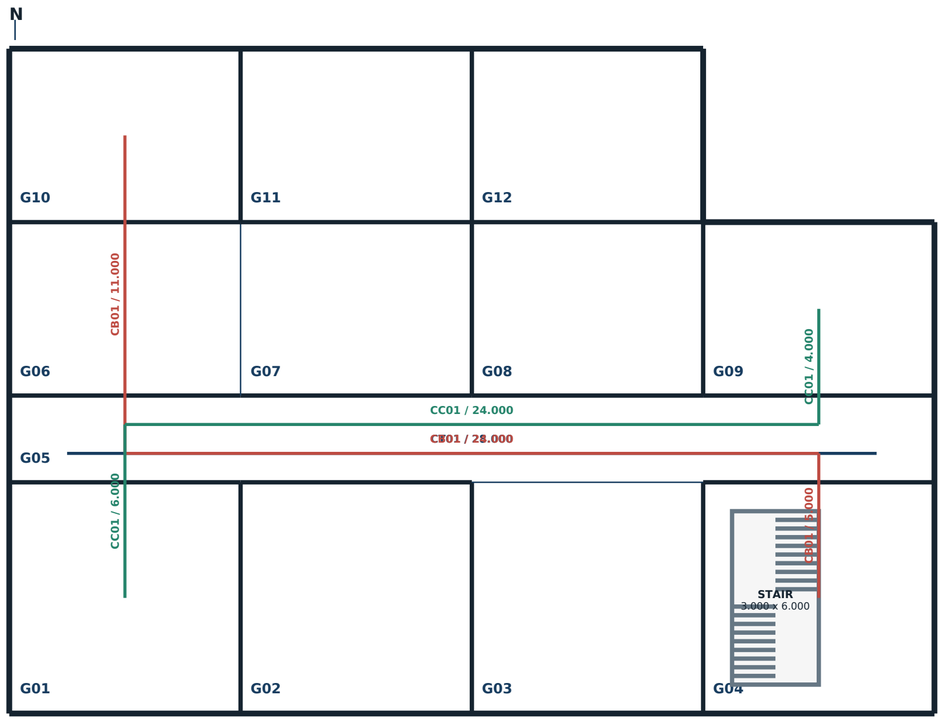
System	Material	Vertical / equipment rule
CW25	PPR cold-water DN25	One 4.200 m interstorey riser on F-CW25 only; zero on G route.
HW20	Insulated PPR hot-water DN20	One 4.200 m interstorey riser on F-HW20 only. Insulation equals HW20 length.
WW50	uPVC waste DN50	One 4.200 m interstorey riser on F-WW50 only.
SW110	uPVC soil DN110	One 4.200 m interstorey riser on F-SW110 only; NO implied vent to roof.
CD25	uPVC condensate DN25	One 4.200 m interstorey riser on F-CD25 only.
WC / WB	WC sets / wash basins	Each of G08 and F08: 4 WC sets + 4 wash basins + 2 floor drains.
SK / FD / WH	Sinks / drains / water heaters	Each G09 and F09: 2 sinks + 1 floor drain + 1 electric water heater.

All pipe diameters are explicitly specified. Do not substitute duct sizes for pipe diameters.

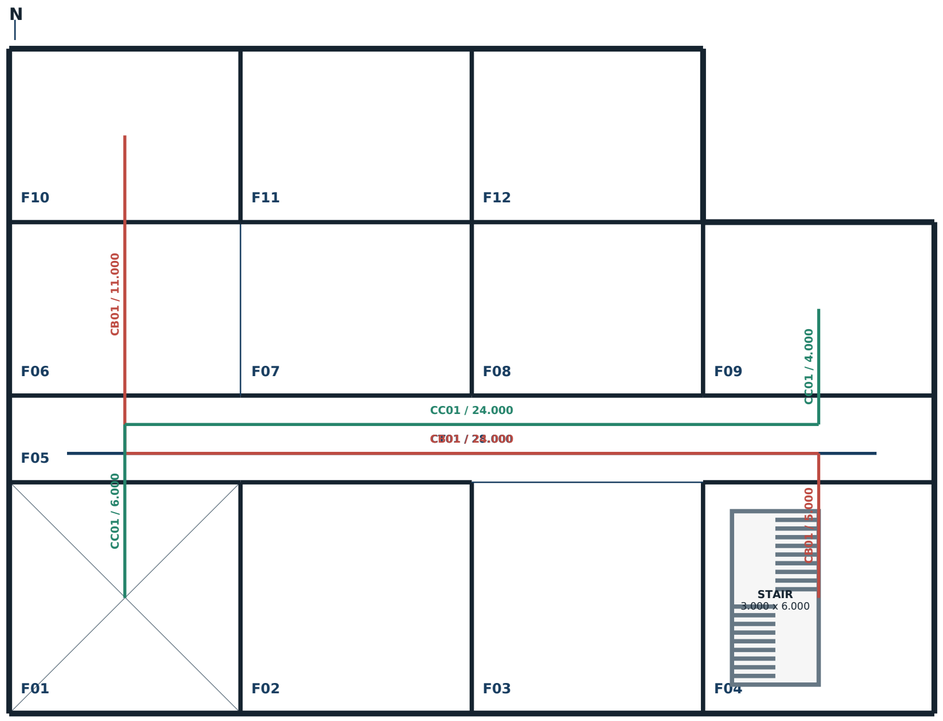
Routes are CL lengths in metres. Ignore apparent crossings; no tee/elbow count inferred. All unshown plumbing is excluded.

Plans 1:150. Equipment matrix is authoritative. Cable routes are limited measured backbone runs, not complete circuit wiring.

GROUND



FIRST



Room	LT01	SO01	DA01	Room	LT01	SO01	DA01
G01	6	4	2	F01	0	0	0
G02	8	6	2	F02	8	10	8
G03	8	6	4	F03	8	6	4
G04	4	2	0	F04	4	2	0
G05	12	4	0	F05	12	4	0
G06	8	10	8	F06	8	8	6
G07	4	4	2	F07	4	4	2
G08	6	2	0	F08	6	2	0
G09	4	6	0	F09	4	6	0
G10	4	4	2	F10	4	4	2
G11	8	8	6	F11	6	6	4
G12	4	2	0	F12	4	2	0

LT01: LED light fittings; SO01: twin socket points;

DA01: data outlets. Matrix entries are counts, not watts.

SW01 light switches: 1 per occupied room except each

corridor has 2. None at F01.

EM01 emergency lights: 3 per corridor + 2 per core.

EX01 exit signs: 3 per level. DB01 distribution board: 1 per level.

CT01: 300 x 50 mm cable tray, one route per level.

CB01: 4C x 16 mm2 power cable, one cable per route.

CC01: 25 mm PVC conduit, one route per level.

Every FIRST route adds one 4.200 m riser; ground adds none.

No extra room wiring, cable slack, switch legs or earthing inferred.

This final sheet defines measurement conventions; it is not a priced BOQ or an expected answer sheet.

Trade	Required result	Common failure to detect
Wall quantities	Separate E01 external and P01 internal net m2	Counting GL01 as CMU; using one height for both levels
Wall finish quantities	Internal plaster, paint, wall tile; exterior paint	Only one face of internal wall; missed clipped tile deductions
Floor finishes	Separate F01 / F02 / F03 / F04 net m2	Counting upper void; adding stair finish twice
Skirting	Separate porcelain / MDF / terrazzo linear metres	Deducting windows; including full glass faces or wet rooms
Screed	SC01 volume m3 at 50 mm	Reporting area as volume; including stair zone
Ceilings	C01 / C02 boards and paint; C03 soffit paint	Counting F01 ceiling again; gypsum at exposed soffits
Doors / windows	Door sets by type; window m2 by type	Same scheduled instance counted twice across sheets
Waterproofing	Wet floor m2 and wall upstand m2 separately	Missing upstand; confusing height with linear measure
Stair / rails	Tread/riser/landing finish areas; rail lengths	Using only rise or flight footprint for finish quantities
HVAC	Duct lengths, sheet area, insulation, equipment counts	Missing explicit 4.8 m riser; counting duct area as floor area
Plumbing	Length per specified pipe diameter; fixture numbers	Missing 4.2 m risers; invented unshown branches
Electrical	Matrix counts + route lengths by specification	Multiplying twin sockets by two; guessed complete wiring

NUMERICAL RULES

Keep full precision during calculation. Report net quantities to 3 decimal places. Round only the final BOQ line.

The geometric reference uses the dimensions in this PDF, not AI-generated object recognition or the application under test.

No waste is included in measured installed quantities. Any purchase waste must be declared separately.

No market rate or fixed project total is specified. Your database rates, markup and VAT determine the money result.

A missing database price should be flagged visibly; it must not erase the measured quantity or invent a zero-cost completion.

For quantity QA compare trade by trade, not only grand total. Different errors can cancel out in a total.

All drawing-sheet repetitions (plans, schedules, details) refer to the SAME work, unless a new instance ID is given.

This is one deterministic stress test. Passing it does not establish accuracy on arbitrary real-world drawings.